

GITHUB WEEK - 2 ASSIGNMENT

Python Program to Find the Factorial of a Number

Problem Statement

Write a Python program to find the factorial of a given number.

1. Objective

To write and execute a Python program that calculates the **factorial of a given number** using a loop.

2. Problem Description

The factorial of a non-negative integer n is the product of all positive integers from 1 to n .

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$$

For example: $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Therefore, when the input is 5, the expected output is: **Factorial of 5 = 120**

3. Algorithm

1. Start the program.
2. Get a number from the user.
3. Initialize the factorial value as 1.
4. Use a for loop from 1 to the given number.
5. Multiply the factorial value by each number in the loop.
6. Display the calculated factorial.
7. Stop the program.

4. Python Program

File Name: factorial.py

```
# Program to find the factorial of a given number

num = int(input("Enter a number: "))

factorial = 1

for i in range(1, num + 1):
    factorial = factorial * i

print("Factorial of", num, "=", factorial)
```

5. Program Explanation

The program first gets an integer from the user using the `input()` function.

```
num = int(input("Enter a number: "))
```

The factorial variable is initialized with 1.

```
factorial = 1
```

A for loop is then used to multiply all the numbers from 1 to the given number.

```
for i in range(1, num + 1):  
    factorial = factorial * i
```

Finally, the calculated factorial is displayed using the `print()` function.

```
print("Factorial of", num, "=", factorial)
```

6. Sample Input

Enter a number: 5

7. Sample Output

Factorial of 5 = 120

8. Output Screenshot

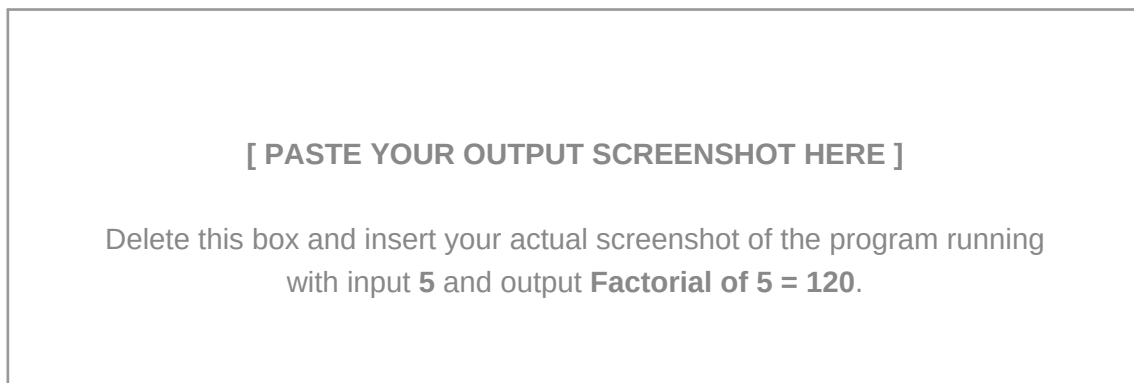


Figure 1: Output of Factorial Python Program

9. GitHub Repository Structure

```
Week-2/  
|-- factorial.py  
|-- README.md  
`-- output.png
```

`factorial.py` - Python source code

`README.md` - Project description and instructions

`output.png` - Screenshot of the program output

10. GitHub Submission Steps

Step 1 - Create the Python file: Create `factorial.py` and paste the program into it.

Step 2 - Execute the program: Run it and enter 5. The output is `Factorial of 5 = 120`.

Step 3 - Take screenshot: Capture the full execution and save it as `output.png`.

Step 4 - Create README: Create `README.md` with the project description.

Step 5 - Upload to GitHub: Upload `factorial.py`, `README.md` and `output.png` to the Week-2 repository.

Git commands:

```
git add factorial.py README.md output.png  
git commit -m "Added Week 2 factorial program"  
git push
```

11. Result

The Python program was successfully implemented and executed to calculate the factorial of a given number. For the input **5**, the program produced **Factorial of 5 = 120**. The Python program, README file, and output screenshot were uploaded to the GitHub repository successfully.

12. GitHub Repository Link

GitHub Repository: [\[Paste your GitHub repository link here\]](#)

Example: <https://github.com/your-username/week-2>